

Due: Thursday February 18th, 2PM or before

----- ★ -----

1. Infinite plane of uniform electric charge.

The x - y plane ($z = 0$) is covered uniformly with electric charge, with surface charge density σ (i.e., any area A of the plane carries charge σA). We will calculate the electric field at some point P away from the plane. Without loss of generality, we can take the point to be $(0, 0, z_0)$, i.e., a point on the z -axis.

- (a) By symmetry, what should be the direction of the electric field $\mathbf{E}$ at the point P ? What should be the direction of the electric field at any other point with positive z -coordinate? What about points with negative z -coordinate?
- (b) Consider an infinitely long strip of the charged plane along the x -direction, running from $x = -\infty$ to $x = +\infty$, parallel to the x -axis. Let the strip be infinitesimally thin, located between y and $y + dy$. Draw this situation. A top view, and a three-dimensional view also showing the z -axis and the point P , might be useful.
- (c) What is the linear charge density of the strip?
- (d) Recall from the previous week: an infinite thin rod of charge, if carrying linear charge density λ , produces a field

$$\frac{1}{2\pi\epsilon_0} \frac{\lambda}{r}$$

at a point which is at radial distance r from the rod. In the present geometry, what is the magnitude and direction of the element of field ($d\mathbf{E}$) produced at P by our infinitesimal strip of charge?

- (e) Find the z -component of $d\mathbf{E}$.
- (f) We now want to use the principle of superposition to obtain the net field at P . This means summing over all infinitesimal strips, i.e., integrating your expression from $y = -\infty$ to $y = \infty$. It might help to remember that

$$\int \frac{du}{u^2 + 1} = \tan^{-1} u \quad \text{and} \quad \lim_{u \rightarrow \pm\infty} \tan^{-1} u = \pm \frac{\pi}{2}$$

- (g) What is the dependence of the net field on z_0 ? Is this surprising?

- (h) You can obtain the same result using Gauss's theorem. For this, you have to devise a closed surface having some surfaces parallel to the plane of charge, and surfaces perpendicular to it. Create such a closed surface. Use Gauss's theorem to derive the electric field at distance z_0 from the infinite plane of charge.

2. Metallic sphere.

Metals are very good electric conductors. An ideal conductor conducts charges perfectly well. In a static situation, the electric potential is constant everywhere in the metal. All electric charges are forced to sit on the surface of the metal. (Otherwise the charges would move until this static situation is reached.)

We consider a solid metallic sphere of radius R , carrying a total charge Q .

- (a) What is the surface charge density on the surface of the sphere?
- (b) What is the electric field at distance $r > R$ from the center of the sphere? Use Gauss's theorem. Do you recognize this expression as the field due to a simpler situation?
- (c) What is the electric field at distance $r < R$ from the center, i.e., within the sphere? (We've mentioned already that the potential is constant at all points of the sphere.)
- (d) What is the electric potential outside the sphere, $r > R$? Use the fact that the electric field is identical to a simpler situation.
- (e) Sketch a plot of the electric potential V as a function of the distance r from the center of the sphere. Make sure to clearly mark the $r = R$ point.